Railway Track Misalignment Detection: Comparing Fine-Tuned ResNet-50, EfficientNet-B3 and MobileNetV3

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Abstract

Misaligned railway tracks are a direct derailment risk, and inspecting them by eye along thousands of kilometres of line is slow and inconsistent. We study how well ImageNet-pre-trained convolutional networks can flag misaligned track from a camera image when only a few hundred labelled examples exist.

*On the TMD SIBGRAPI-2021 track-misalignment dataset (555 images, two classes) we fine-tune three architectures that span two orders of magnitude in size: ResNet-50, EfficientNet-B3 and MobileNetV3-Large. All three share one training recipe: partial unfreezing, AdamW, label smoothing, heavy augmentation and early stopping on a stratified validation split. ResNet-50 performs best with **96.4% test accuracy** and balanced errors (2 missed defects, 2 false alarms). EfficientNet-B3 reaches 93.7%, and MobileNetV3 reaches 91.9% with no false alarms but 9 missed defects. We discuss what these error profiles mean for a safety-critical deployment and how much confidence a 111-image test set supports.*

1. Introduction

The real-world problem. Track geometry defects, such as rails that have shifted laterally or buckled, build up gradually and are found today mostly by inspection trains or track walkers. An image classifier mounted on regular rolling stock could screen far more track, far more often, and send only suspicious frames to a human.

The data challenge. Labelled defect images are rare. The dataset used here has 444 training images, which is far too few to train a modern CNN from scratch. Transfer learning [8] addresses this: features learned on ImageNet [1] (edges, textures, parts) are reused, and only the higher layers are adapted to track imagery.

What we compare. We fine-tune three widely used backbones with very different capacity: ResNet-50 [2] (23.5 M parameters), EfficientNet-B3 [6] (11.1 M) and MobileNetV3-Large [3] (3.2 M). We report accuracy and per-class error profiles under a shared recipe.

Table 1. Backbones and fine-tuning scope.

Model	Trainable	Total	Share
ResNet-50	22.07 M	23.51 M	94%
EfficientNet-B3	8.90 M	11.09 M	80%
MobileNetV3-Large	2.00 M	3.22 M	62%

2. Related Work

Transfer learning. Yosinski *et al.* [8] showed that early convolutional features are general while later ones are task-specific, which motivates freezing early blocks and fine-tuning late ones when data are scarce.

Backbones. ResNet [2] uses residual connections, which keep gradients stable in deep networks and make it a standard transfer-learning baseline. EfficientNet [6] scales depth, width and resolution jointly for accuracy per FLOP. MobileNetV3 [3] combines depthwise-separable convolutions, squeeze-and-excitation and hard-swish activations for on-device inference.

Regularisation for small data. Label smoothing [5] discourages over-confident predictions, and decoupled weight decay (AdamW) [4] gives cleaner regularisation with adaptive optimisers.

3. Method

3.1. Fine-tuning strategy

Each backbone is initialised from ImageNet weights. Early stages stay frozen to preserve generic filters; the last stages and a new task head are trained (Table 1).

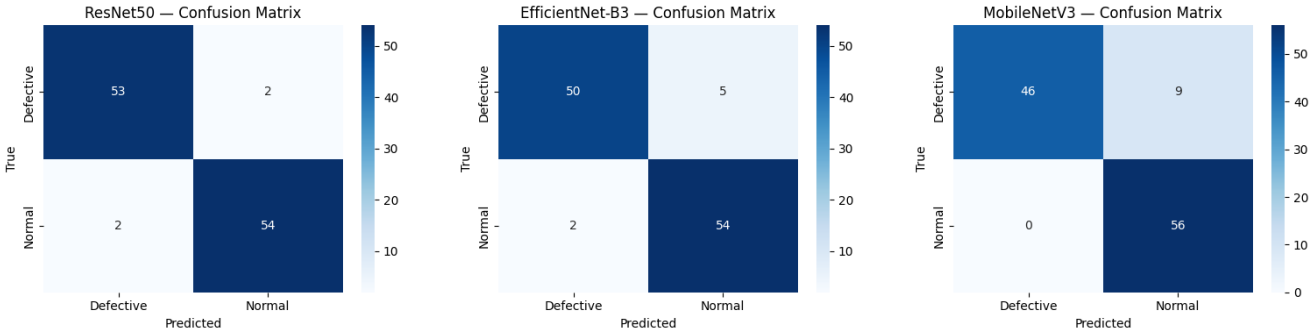
- **ResNet-50:** `layer3`, `layer4` and a new Dropout–Linear head are trainable.
- **EfficientNet-B3** (timm): the last two MBConv stages plus a head BN → Dropout → Linear(256) → ReLU → Dropout → Linear(2).
- **MobileNetV3-Large:** the last three feature blocks plus a head Linear(256) → Hard-swish → Dropout → Linear(2).

3.2. Training recipe

All models share one recipe so that differences come from the architecture: AdamW (learning rate 3×10^{-4}, weight decay 10^{-4}) on trainable parameters only; cross-entropy

Table 2. Test-set results on 111 images. Precision (P) and recall (R) per class; best accuracy in bold.

Model	Test acc.	Defective P	Defective R	Normal P	Normal R	Macro F1	Errors (FN / FP)
ResNet-50	**96.40%**	0.96	0.96	0.96	0.96	**0.96**	4 (2 / 2)
EfficientNet-B3	93.69%	0.96	0.91	0.92	0.96	0.94	7 (5 / 2)
MobileNetV3-Large	91.89%	1.00	0.84	0.86	1.00	0.92	9 (9 / 0)



(a) ResNet-50 (b) EfficientNet-B3 (c) MobileNetV3-Large

Figure 1. Test-set confusion matrices (rows: true class; columns: predicted class).

Table 3. Dataset splits.

Split	Defective	Normal	Total
Train	188	190	378
Validation (15% of train)	33	33	66
Test	55	56	111
Total	276	279	555

with label smoothing 0.1; ReduceLROnPlateau on validation accuracy (factor 0.5, patience 5); batch size 16; at most 30 epochs with early stopping (patience 10); head dropout 0.4; gradient clipping at norm 1.0; and 224×224 inputs with ImageNet normalisation. The checkpoint with the best validation accuracy is restored before testing.

Data augmentation. With so few images, augmentation is the main defence against overfitting. Training images are resized to 256 and randomly cropped to 224, flipped horizontally ($p = 0.5$) and vertically ($p = 0.3$), rotated by up to $\pm 20^\circ$, colour-jittered (brightness and contrast 0.4, saturation 0.3, hue 0.05) and occasionally converted to grayscale ($p = 0.05$). These transforms mimic changes in camera angle and lighting. Validation and test images are only resized and normalised.

4. Results and Discussion

4.1. The data

The dataset [7] provides train and test folders of *Defective* (misaligned) and *Normal* track images. We carve a stratified 15% validation split from each training class with a fixed seed (Table 3). Classes are balanced in every split.

4.2. Quantitative comparison

Table 2 reports test-set performance and Figure 1 the confusion matrices. Here FN is a defective track predicted normal (a missed defect) and FP is a normal track flagged as defective (a false alarm).

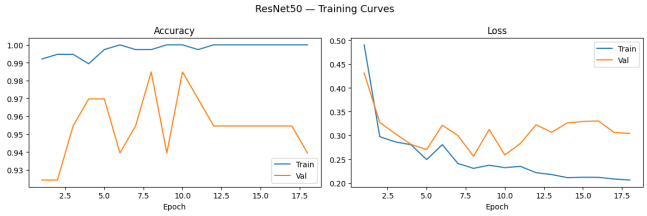
ResNet-50 is the most accurate. It makes only four errors, split evenly between missed defects and false alarms. It also has by far the most trainable parameters (22.1 M). On this dataset, more adaptable capacity in the late stages helped rather than overfitted, probably because augmentation and label smoothing kept it in check.

EfficientNet-B3 is close but unstable. Its validation loss shows a large spike around epoch 8 (Fig. 2b). A plausible cause is the BatchNorm1d in its custom head, which estimates statistics from batches of only 16 samples; a larger batch or LayerNorm in the head would test this.

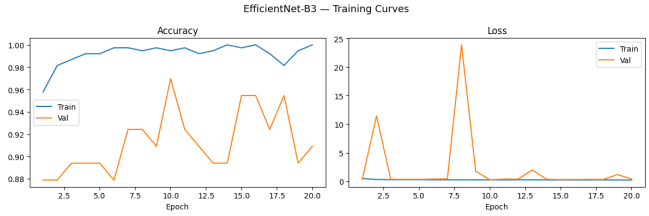
MobileNetV3 never raises a false alarm but misses 16% of defects. In a safety application a missed defect costs far more than an unnecessary inspection, so this model would need its decision threshold lowered before deployment. Its precision of 1.00 on defects suggests there is room to do so.

4.3. Training behaviour

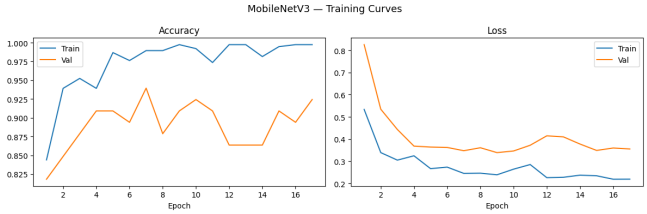
All models reach near-100% training accuracy within about 8 epochs, while validation accuracy plateaus at 90–97% (Fig. 2). The pre-trained features clearly transfer, and the limit is the amount of data rather than model capacity. Early stopping ended training after roughly 18 (ResNet-50), 20 (EfficientNet-B3) and 17 (MobileNetV3) epochs.



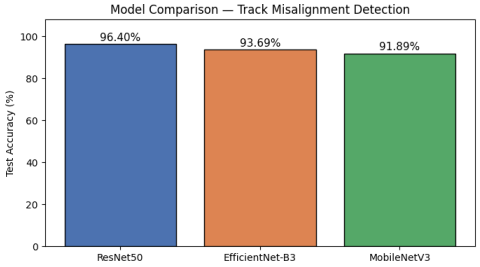
(a) ResNet-50



(b) EfficientNet-B3



(c) MobileNetV3-Large



(d) Test accuracy of the three models

Figure 2. (a)–(c) Training and validation accuracy (left panel) and loss (right panel) per backbone. (d) Final test accuracy.

4.4. How much should we trust the ranking?

With 111 test images, a single image moves accuracy by 0.9 points, and the 95% binomial interval around 96% is roughly ± 3.5 points. The ordering ResNet-50 $>$ EfficientNet-B3 $>$ MobileNetV3 is therefore suggestive, not conclusive. Repeated runs with different seeds, or k-fold cross-validation over the 555 images, would be needed to separate models that differ by 2–3 points.

5. Conclusion & Future Work

Partial fine-tuning of ImageNet backbones detects track misalignment accurately from only 378 training images: ResNet-50 reaches 96.4% test accuracy with balanced errors. Lighter networks trade accuracy for speed; MobileNetV3 is conservative, never raising a false alarm but missing more defects. **Next steps.** (i) Choose operating points from ROC curves so that defect recall meets a safety target; (ii) estimate variance with cross-validation or repeated seeds; (iii) add Grad-CAM maps to confirm the networks look at the rails rather than background cues; and (iv) benchmark inference latency on embedded hardware, where MobileNetV3's size would matter.

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